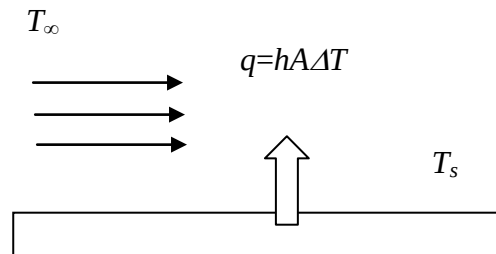


6. EXTENDED SURFACES

<Motivation - heat transfer enhancement>

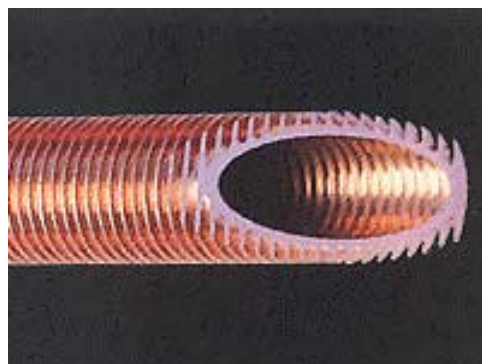
How to enhance HT? For example, how to design a chip-cooling system?



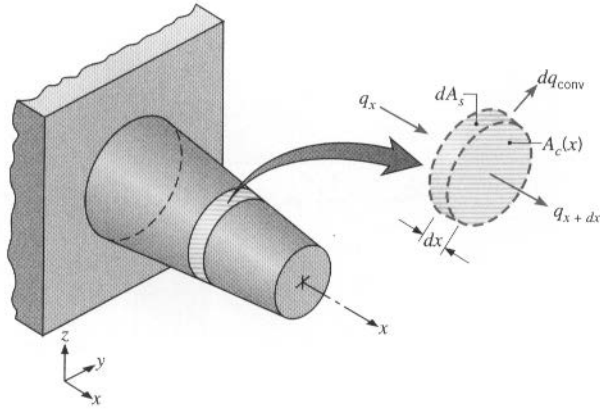
- Increase h
- Increase A
- Increase ΔT

- Increasing heat transfer coefficient
 - Gas flow → liquid flow
 - Natural convection → Forced convection (increased velocity)
 - Single phase flow → Two phase flow (phase change HT)
 - Laminar flow → turbulent flow
 -
- Increasing contact area
 - Extended surfaces or fins

(1) Fins



(2) Deriving a governing equation for a single fin



$$q_x + dq_{conv} = q_{x+dx}$$

$$q_x = -kA_c \frac{dT}{dx}$$

$$q_{x+dx} = q_x + \frac{dq_x}{dx} \delta x$$

$$dq_{conv} = hPdx(T - T_\infty) \quad (P: \text{perimeter, e.g., } P = \pi D)$$

$$\boxed{\frac{d^2T}{dx^2} - \frac{hP}{kA_c}(T - T_\infty) = 0} \quad \text{for steady state, constant cross-sectional area}$$

(3) Solutions to fin equations

$$\frac{d^2\theta}{dx^2} - m^2\theta = 0 \quad \text{where } \theta \equiv T - T_\infty, \quad m^2 = \frac{hP}{kA_c}$$

B.C. (infinitely long fin)

- $\theta(0) = T_b - T_\infty = \theta_b$
- $\theta(\infty) = T_\infty - T_\infty = 0$

$$\theta = C_1 e^{mx} + C_2 e^{-mx}$$

$$\text{B.C.} \rightarrow C_1 = 0 \quad \text{and} \quad C_2 = \theta_b$$

$$\therefore \boxed{\theta = \theta_b e^{-mx}}$$

$$\text{Fin heat transfer rate } q_f = -kA_c \left. \frac{d\theta}{dx} \right|_{x=0} = \sqrt{hPkA_c} \theta_b$$

Table 3.4 Incropera

TABLE 3.4 Temperature distribution and heat loss for fins of uniform cross section

Case	Tip Condition ($x = L$)	Temperature Distribution θ/θ_b	Fin Heat Transfer Rate q_f
A	Convection heat transfer: $h\theta(L) = -k d\theta/dx _{x=L}$	$\frac{\cosh m(L-x) + (h/mk) \sinh m(L-x)}{\cosh mL + (h/mk) \sinh mL}$ (3.70)	$M \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL}$ (3.72)
B	Adiabatic $d\theta/dx _{x=L} = 0$	$\frac{\cosh m(L-x)}{\cosh mL}$ (3.75)	$M \tanh mL$ (3.76)
C	Prescribed temperature: $\theta(L) = \theta_L$	$\frac{(\theta_L/\theta_b) \sinh mx + \sinh m(L-x)}{\sinh mL}$ (3.77)	$M \frac{(\cosh mL - \theta_L/\theta_b)}{\sinh mL}$ (3.78)
D	Infinite fin ($L \rightarrow \infty$): $\theta(L) = 0$	e^{-mx} (3.79)	M (3.80)
$\theta \equiv T - T_\infty$ $m^2 \equiv hP/kA_c$ $\theta_b = \theta(0) = T_b - T_\infty$ $M \equiv \sqrt{hPkA_c} \theta_b$			

(4) Fin performance

- Fin resistance: $R_{t,f} = \frac{T_b - T_\infty}{q_f}$
- Fin effectiveness: $\varepsilon_f = \frac{q_f}{q_{\text{without a fin}}} = \frac{q_f}{hA_{sb}(T_b - T_\infty)} = (\text{heat transfer with a fin}) / (\text{heat transfer without a fin})$
- Fin efficiency: $\eta_f = \frac{q_f}{q_{\text{max}}} = \frac{q_f}{hA_f(T_b - T_\infty)} = (\text{heat transfer with a fin}) / (\text{heat transfer with an ideal fin})$